

Diagnosing Separation and Initial-Condition Errors

AP Calculus AB/BC • Error Analysis

Directions. Separable equations go wrong in three predictable places: integrating before the variables are actually separated, losing the constant of integration, and applying the initial condition at the wrong moment. Where a student's work is shown, **(i)** name the first wrong line, **(ii)** say what rule it breaks, **(iii)** solve correctly. Round decimals to two places and check each solution by substituting it back into the differential equation.

- 1.** Find the general solution of $\frac{dy}{dx} = 6x^2 - 4x$. Include the constant of integration, and say in one phrase what it represents on a slope field.

Answer: _____

- 2.** Same equation, $\frac{dy}{dx} = 6x^2 - 4x$, now with $y(1) = 5$. Find the particular solution and evaluate $y(2)$.

Answer: _____

3. Find and fix. Solve $\frac{dy}{dx} = xy$ with $y(0) = 2$; find $y(2)$. Wren's attempt:

$$\int dy = \int xy \, dx \Rightarrow y = \frac{x^2}{2} + C; \quad y(0) = 2 \Rightarrow C = 2; \quad y(2) = 2 + 2 = 4$$

Name the first wrong line, say what Wren did to the y on the right side, and give the correct $y(2)$.

Answer: _____

4. Find and fix. Solve $\frac{dy}{dx} = 3y$ with $y(0) = 4$; find $y(1)$. Dev's attempt:

$$\ln |y| = 3x + C \Rightarrow y = e^{3x} + C; \quad y(0) = 4 \Rightarrow C = 3; \quad y(1) = e^3 + 3 \approx 23.09$$

Dev's first line is correct. Name the line that is not, explain what exponentiating does to a sum, and find $y(1)$ correctly.

Answer: _____

5. Find and fix. Solve $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = 3$; find $y(4)$. Rosa separates correctly to $\frac{y^2}{2} = \frac{x^2}{2} + C$, then writes:

“Square-root both sides”: $y = x + C$; $y(0) = 3 \Rightarrow C = 3$; $y(4) = 4 + 3 = 7$

Two things went wrong in that first line. Name both, then solve correctly for $y(4)$.

Answer: _____

6. Find and fix. Solve $\frac{dy}{dx} = y^2$ with $y(0) = 1$; find $y(0.5)$. Sam’s attempt:

$\int y^{-2} dy = \int dx \Rightarrow \frac{1}{y} = x + C$; $y(0) = 1 \Rightarrow C = 1$; $y(0.5) \approx 0.67$

One sign is missing. Fix it and give $y(0.5)$. Then say what the corrected solution does as x approaches 1, and why that behaviour is invisible in Sam’s version.

Answer: _____

7. Solve $\frac{dy}{dx} = \frac{2x}{x^2 + 1}$ with $y(0) = 1$ and evaluate $y(2)$ to two decimal places. Then state the step at which the initial condition must be applied, and why applying it before integrating would be meaningless.

Answer: _____

8. A classmate argues: “As long as you add a constant somewhere, it does not matter whether you add it before or after exponentiating.” Using $\frac{dy}{dx} = 3y$ from problem 4, show this is false: differentiate $y = e^{3x} + 3$, compare the result with $3y$, and explain what the constant becomes when it is carried through the exponential correctly.